

PANEL DATA

Introduction and Why Study Panel Data

Class 1 · Hour 1

Prof. Dr. Rodrigo Leite
Panel Data · EBAPE/FGV

CLASS 1 · HOUR 1

What We'll Cover This Hour

01

Who I am and how the course works

Research background, the nine-class roadmap, assessment, and the rhythm of each meeting

02

Ground rules

The one rule that matters most: interrupt me

03

Why panel data?

Why this is one of the most flexible tools in empirical research

04

Panel data across fields

Behavioral science, strategy, policy, and finance: panels are everywhere

About the Professor



Most of my research is motivated by questions in **financial economics**, and it also draws on accounting, psychology, and public policy.

Research interests

- How people make financial decisions, and how behavioral biases shape individual and group performance
- Financial institutions and disadvantaged groups: microfinance and microcredit
- Efficiency of public procurement and state-owned enterprises

Selected outlets

Journal of Public Administration Research and Theory · Journal of Business Ethics · The International Journal of Accounting · Accounting Forum · International Journal of Finance and Economics · Finance Research Letters

Course Roadmap: Nine Classes

Class	Date	Theme	Topics
1	Oct 27	**The Basics of Panel Data**	Why panel data; panel structure; pooled OLS
2	Oct 30	**Panel Data Regressions**	RE vs. FE; Hausman & Breusch–Pagan tests
3	Nov 3	**Advanced Panel Data Models I**	Pseudo-panels; panel IV regressions
4	Nov 10	**Advanced Panel Data Models II**	Diff-in-diff; dynamic panels and GMM
5	Nov 17	**Midterm Exam**	Hours 1–3
6	Nov 24	**Hands-On**	Midterm discussion; Stata exercise
7	Dec 1	**Extensions I: Time Series**	AR, MA, ARMA, ARIMA; ARCH and GARCH
8	Dec 8	**Extensions II: Bayesian Models**	Bayesian models; Bayesian regressions
9	Dec 15	**AI & Mock-Up Conference**	AI in research; project presentations

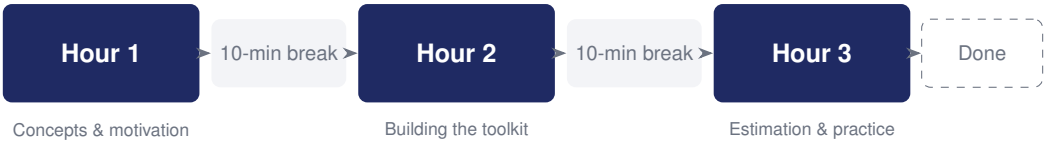
Hour 3 of Classes 2–4 and 6–8: paper presentations. **Deadlines:** hands-on assignment in Class 8; manuscript by **Dec 8, noon**.

How You Will Be Assessed

Component	Weight	Timing
Midterm exam (closed book)	30%	Class 5 (Nov 17)
Assignments (two)	30%	A hands-on data assignment, due in Class 8; a review of a real paper, due two weeks after I send it
Research project presentation	30%	Class 9 (Dec 15): mock conference, 15–20 min + 5–10 min of discussion; manuscript due Dec 8, noon
Paper presentations	10%	Classes 2–4 and 6–8, three papers per class



How Each Meeting Is Structured



Each hour opens with a short agenda and closes with a recap and a preview of what comes next, so you always know where we are.

The One Rule That Matters Most

Always — and I mean **always** — interrupt me to ask questions.

Concepts stack

You need each concept to understand the next one. If you are stuck now, you will stay stuck.

It's my job

Part of my job is to make sure everyone understands a concept before we move on.

Five minutes is cheap

Better to pause for five minutes than to have part of the room tune out.

Why I Like Teaching This Course

Panel data is one of the most **flexible** estimation frameworks you can learn. It shows up whenever you:

Run experiments in waves

Same subjects observed before and after, or across several rounds

Exploit natural experiments

DiD, RDD, RD-in-time, event studies

Track relationships over time

Correlations that evolve as units change

Go further

Dynamic panels (GMM), panel IV, and beyond

> Most of the papers I have published use panel data in some capacity.

“I Won’t Use Panel Data in My Research”

“That’s fine for finance and economics, but it’s not relevant to my line of research.”

Before ruling it out, consider:

1	**Almost every dataset has a hidden time dimension**	Repeated surveys, sequential transactions, yearly filings: the “when” is usually already there, just unused.
2	**Ignoring it doesn’t remove the problem**	Skipping the panel structure leaves unobserved heterogeneity and dynamics unaddressed, not solved.
3	**The logic transfers across designs**	Repeated-measures experiments, event studies, and diff-in-diff all rely on the same panel intuition.
4	**Almost any repeated-unit question qualifies**	Comparing the same people, firms, or places at more than one moment already makes it a panel.

Panel Data in Behavioral & Decision Sciences

Repeated experimental sessions	The same subjects deciding across rounds or treatments: a within-subject panel
Longitudinal well-being surveys	The same respondents' life satisfaction, risk preferences, and happiness tracked over years
Cognitive ability and careers	Individuals followed from cognitive-reflection tests into repeated labor-market outcomes
Consumer sentiment panels	The same households' choices and attitudes, wave after wave
Learning-curve studies	The same decision-makers observed as they accumulate experience

Panel Data in Strategy, Management & Organizations

Firm strategy panels	The same firms over years: diversification, M&A activity, performance persistence
Executive and board panels	CEO turnover, board composition, and compensation, year after year
Innovation panels	Patenting and R&D activity of the same firms over time
Practice-adoption panels	Which firms adopt governance codes or ESG practices, and when
Corporate social performance	State ownership, political ideology, and social performance across firms and years

Panel Data in Institutions, Policy & Government

Institutional-quality panels	Governance indicators across countries over decades
State-owned enterprise panels	SOEs across countries and years: political ideology and performance
Public procurement panels	Government contracts across municipalities and years
Policy diffusion panels	Which regions adopt a reform, and when
Political-economy panels	Government spending and election cycles across jurisdictions and years

Panel Data in Finance

Firm-level financial panels	Capital structure, governance, and performance across firms and years
Banking panels	Cash holdings, lending behavior, and systemic risk across banks over time
Asset-pricing panels	Stock returns across firms and periods, as in the Fama–French factor models
Microfinance institution panels	Profit orientation and earnings management across institutions and years
Sovereign-debt panels	Debt renegotiation outcomes across borrowers and time

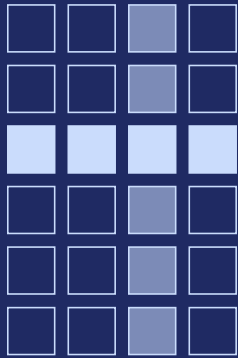
Recap and What's Next

Takeaway: panel data is not just for economists. Whenever you observe the *same units* more than once, you have a panel, and ignoring that structure has consequences.

Next: Hour 2 — Panel Data Structure

What exactly a panel is, how it differs from a cross-section and a time series, and the four reasons researchers use it.

Questions?



PANEL DATA

Panel Data Structure

Class 1 · Hour 2

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CLASS 1 · HOUR 2

What We'll Cover This Hour

01 **What panel data is**
The two dimensions every panel combines: units and time

02 **How panel data is structured**
Notation, long vs. wide format, balanced vs. unbalanced, short vs. long

03 **Panel vs. cross-section vs. time series**
And why a repeated cross-section is *not* a panel

04 **Why researchers use panel data**
Four core motivations for the method

05 **What panel data can't do**
Honest limits, before we start estimating

What Is Panel Data?

Panel (or **longitudinal**) data follows the *same* cross-sectional units — firms, individuals, countries, banks — repeatedly over multiple time periods.

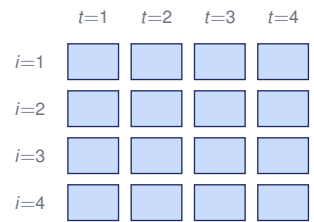
It merges two dimensions:

Cross-section: the *i* dimension

Who — firms, people, states

Time series: the *t* dimension

When — years, quarters, months



Each cell: one unit i observed at one period t

The Anatomy of a Panel: Notation

$$y_{it} = \beta_0 + \beta_1 x_{it} + \alpha_i + \varepsilon_{it}, \qquad i = 1, \ldots, N; \quad t = 1, \ldots, T$$

i	**unit index**	firm, individual, country, bank (N units)
t	**time index**	year, quarter, month (T periods)
α_i	**unit effect**	everything time-invariant and unobserved about unit i (no t subscript!)
ε_{it}	**idiosyncratic error**	the leftover shock specific to unit i at time t

A balanced panel has $N \times T$ observations; an unbalanced one has $\sum_i T_i$.

How a Panel Looks in Your Data File: Long vs. Wide

Long format (one row per *i*, *t*)

id	year	y	x
1	2019	4.1	0.3
1	2020	4.4	0.5
2	2019	3.2	0.1
2	2020	3.0	0.2

What `xtset` and almost every panel estimator expect

Wide format (one row per *i*)

id	y2019	y2020	x2019	x2020
1	4.1	4.4	0.3	0.5
2	3.2	3.0	0.1	0.2

Common in spreadsheets and survey releases

```
reshape long y x, i(id) j(year)     // wide -> long
reshape wide y x, i(id) j(year)     // long -> wide
```

Values are illustrative.

Not All Panels Look Alike

Balanced

Every unit observed in every period: a complete $N \times T$ rectangle.

e.g., all 27 Brazilian states, 2000–2020

Unbalanced

Units enter, exit, or have missing periods.

e.g., firms IPO, delist, or miss a filing year

Short panel (large N, small T)

Thousands of firms, a handful of years.

Asymptotics: $N \to \infty$, T fixed

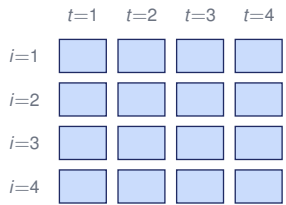
Long panel (small N, large T)

A few countries tracked over decades.

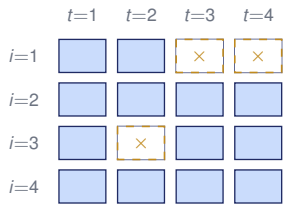
Asymptotics: $T \to \infty$; time-series issues return

Most panels in business research are **short and unbalanced** — this shapes which estimators and standard errors we use.

Balanced vs. Unbalanced Panels



Balanced: all $N \times T = 16$ cells observed



Unbalanced: missing $(i, t) = (1, 3),\ (1, 4),\ (3, 2)$

Ask *why* cells are missing. Unit 1 disappearing after $t = 2$ could be a merger, a bankruptcy, or a survey drop-out — and **non-random attrition** can bias your estimates.

A Common Confusion: Panel vs. Repeated Cross-Sections

Repeated (pooled) cross-sections

- — A *new* random sample is drawn each period
- — You cannot follow the same unit over time
- — You can study how the population changes, not how individuals change

Panel data

- \+ The *same* units are re-observed each period
- \+ Each unit carries a persistent identifier (i)
- \+ You can study within-unit change and net out α_i

The test: **can you link observation** (i, t) **to observation** $(i, t+1)$**?** If not, it is not a panel — no matter how many years it covers.

Panel Data vs. Pure Cross-Section

A cross-section gives one snapshot; a panel observes the same units again and again.

Cross-section only

- — One observation per unit, one moment in time
- — Cannot separate a unit's fixed traits from the effect of *x*
- — No way to track how a unit changes over time
- — Unobserved heterogeneity stays in the error term

Panel data

- \+ Multiple observations per unit, across time
- \+ Can net out each unit's stable, unobserved traits
- \+ Tracks within-unit change, not just cross-unit differences
- \+ Lets you control for unobserved heterogeneity

Panel Data vs. Pure Time Series

A time series tracks one unit over time; a panel adds many units observed the same way.

Time series only

- — A single unit followed across many periods
- — Findings describe that one unit: limited generalizability
- — Needs long T for reliable inference; often nonstationary
- — Cannot exploit variation across units

Panel data

- \+ Many units, each followed over time
- \+ Findings generalize to the sampled population
- \+ Cross-unit variation adds power even with short T
- \+ Combines within-unit dynamics and cross-unit differences

Why Do Researchers Use Panel Data? Four Reasons

1

Control unobserved heterogeneity

Net out stable traits that would otherwise bias your estimates

2

Study dynamics

Separate short-run adjustment from long-run effects

3

Richer, more informative data

More variation, less collinearity, more efficient estimates

4

Detect hidden effects

Identify relationships neither dimension reveals alone

See Baltagi (2021, ch. 1) and Hsiao (2022, ch. 1) for extended discussions.

Reason 1: Controlling Unobserved Heterogeneity

Every firm, person, or country has stable traits we cannot measure: management quality, culture, innate ability, institutional strength.

Without panel structure

One period only: α_i hides in the error.

$$y_i = \beta_0 + \beta_1 x_i + \underbrace{\alpha_i + \varepsilon_i}_{u_i}$$

If $\text{Cov}(x_i, \alpha_i) \neq 0$, OLS is **biased and inconsistent**.

With panel structure

Difference two periods of the same unit:

$$\Delta y_{it} = \beta_1 \Delta x_{it} + \Delta \varepsilon_{it}$$

α_i **cancels out** — whatever it is, however large.

First differences are one transformation; **fixed effects** (next class) are another.

Reason 2: Studying Dynamics of Adjustment

A cross-section shows where units end up. A panel shows the path they took to get there.

$$y_{it} = \rho\, y_{i,t-1} + \beta x_{it} + \alpha_i + \varepsilon_{it}, \qquad |\rho| < 1$$

1. **Short-run response** — The immediate reaction to a shock: $\partial y_{it}/\partial x_{it} = \beta$
2. **Long-run equilibrium** — Where the outcome settles once adjustment finishes: $\beta/(1-\rho)$
3. **Speed of adjustment** — How quickly units close the gap each period: $1-\rho$

Example: after a governance reform, does firm performance jump immediately, drift up over several years, or both?

Reason 3: More Data, More Efficient Estimates

$N \times T$

observations instead of just N or just T

↓

collinearity among regressors, as more variation is exposed

↑

degrees of freedom, sharpening the precision of estimates

↑

efficiency relative to a single cross-section or time series

Bottom line: combining cross-sectional and time variation gives you more identifying variation than either dimension alone.

Reason 4: Seeing What Neither Dimension Reveals Alone

Some questions can only be answered by combining within-unit and across-unit variation.

Cross-section alone	confuses differences *across* firms with the effect you actually want to estimate
Time series alone	confuses one firm's history with a general economic relationship
Panel data	separates within-unit change from across-unit differences, isolating the effect of interest

Panel Data in Practice

Panel structures are everywhere in empirical finance, business, and public administration:

Firm-level financial panels	Listed-firm data (e.g., Compustat): governance, performance, capital structure over years
State-owned enterprise panels	SOEs across countries and years: political ideology and performance
Banking and credit panels	Bank balance sheets over time: cash holdings, risk, deforestation-linked lending
Public procurement panels	Government contracts across municipalities and years: policy uncertainty

Several of these come from papers we will present and discuss later in the course.

Panel Data Is Powerful, But It Is Not Magic

Attrition

Units drop out non-randomly, which can bias the remaining sample.

Short-T problems

Several estimators behave poorly when the time dimension is very short (e.g., dynamic panels with fixed effects).

Endogeneity persists

Fixed effects remove *time-invariant* confounders — not time-varying ones, reverse causality, or measurement error.

Data demands

Requires repeated, comparable measurement, which is harder to collect than a single survey.

Check Your Understanding: Is It a Panel?

(a) An annual household survey that draws a new random sample every year

(b) Financial statements of the same listed firms, 2010–2024

(c) A lab experiment in which each subject plays 20 rounds

(d) Monthly inflation for Brazil, 1995–2024

Check Your Understanding: Is It a Panel?

(a) An annual household survey that draws a new random sample every year	**Repeated cross-section**
(b) Financial statements of the same listed firms, 2010–2024	**Panel (likely unbalanced)**
(c) A lab experiment in which each subject plays 20 rounds	**Panel (subject $\times$ round)**
(d) Monthly inflation for Brazil, 1995–2024	**Time series ($N = 1$)**

The deciding question is always the same: **are the same units observed more than once, and can you link them?**

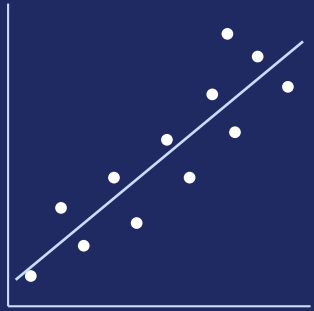
Recap and What's Next

Takeaway: panel data combines the cross-section and the time series, letting us control for unobserved heterogeneity, study dynamics, and estimate more efficiently — but it does not solve every identification problem on its own.

Next: Hour 3 — Pooled OLS

Our first panel estimator: what it does, when it works, and how to get its standard errors right.

Questions?



POOLED OLS

Pooled OLS Regressions

Class 1 · Hour 3

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CLASS 1 · HOUR 3

What We'll Cover This Hour

01 **The pooled OLS model**
Stacking every unit-period into one big regression

02 **Assumptions for consistency**
What has to be true for pooled OLS to be believable

03 **The standard-error problem**
Why naive SEs mislead, and how clustering fixes it

04 **Testing poolability and reading the output**
A formal test, and between vs. within variation

05 **Estimating it in Stata**
Syntax, output, and how to read the coefficients

What Is Pooled OLS?

Pooled OLS is the simplest way to estimate a panel model: stack every unit-period observation into one big sample and run ordinary least squares *as if the panel structure didn't exist*.

It ignores two things a panel gives you for free

1. Observations from the same unit *i* are **not independent** of each other.
2. Each unit may have its own **stable, unobserved traits** (α_i).

Think of it as: *$N \times T$* ***observations, one single regression line.***

The Pooled OLS Equation

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it}, \qquad u_{it} = \alpha_i + \varepsilon_{it}$$

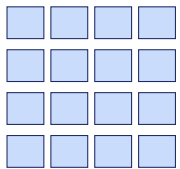
Notice what's missing:

There is **no separate α_i term**. The unit effect is folded into the *composite error* u_{it}.

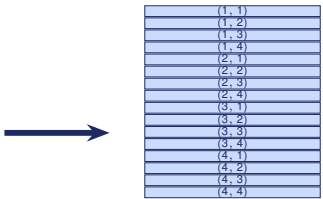
This is exactly why pooled OLS lives or dies on whether that unit effect is a problem.

From a Panel to One Big Sample

Pooled OLS collapses the $i \times t$ grid into a single column of $N \times T$ rows; the panel structure disappears from the estimator's point of view.



Panel ($i \times t$)



Stacked sample ($N \times T$ rows)

One regression, no unit index, no time index in the estimator itself — just $N \times T$ pairs (x, y).

Assumptions Pooled OLS Needs

1	**Linearity**	The model is linear in parameters: $y_{it} = \boldsymbol{x}_{it}\beta + u_{it}$
2	**Contemporaneous exogeneity**	$E(\boldsymbol{x}_{it}' u_{it}) = \mathbf{0}$ for every t: the regressors are uncorrelated with the composite error *in the same period*
3	**Rank condition**	No perfect collinearity: $\operatorname{rank} E\left(\sum_t \boldsymbol{x}_{it}' \boldsymbol{x}_{it}\right) = K$
4	**For classical SEs only**	Homoskedasticity **and** no serial correlation in u_{it} — rarely true in panels, hence clustering

Pooled OLS does *not* need **strict** exogeneity, $E(u_{it} \mid \boldsymbol{x}_{i1}, \ldots, \boldsymbol{x}_{iT}) = 0$. That stronger condition will matter for fixed and random effects next class. (Wooldridge, 2010, ch. 7)

The Critical Assumption: No Hidden Heterogeneity

Since $u_{it} = \alpha_i + \varepsilon_{it}$, assumption 2 requires that the unit effect buried in the error is uncorrelated with x_{it}.

If $\text{Cov}(x_{it}, \alpha_i) = 0$

Pooled OLS is **consistent**: the unit effect is just noise, uncorrelated with the regressor of interest.

(Standard errors still need fixing — see below.)

If $\text{Cov}(x_{it}, \alpha_i) \neq 0$

Pooled OLS is **biased and inconsistent**: the omitted unit effect leaks into $\hat{\beta}$.

More data does not help — the bias does not shrink as N grows.

A Concrete Example: Omitted Variable Bias

Question: does board independence raise firm profitability?

The omitted variable	Management quality (α_i): stable, unobserved, varies by firm
The correlation problem	Better-managed firms may also choose more independent boards: $\text{Cov}(x_{it}, \alpha_i) > 0$
The consequence	The estimate partly reflects management quality, not governance itself

With a single regressor: $\text{plim}\ \hat{\beta}_1^{POLS} = \beta_1 + \dfrac{\text{Cov}(x_{it}, \alpha_i)}{\text{Var}(x_{it})}$ $\Rightarrow$ **upward bias** here.

When Pooled OLS Is a Reasonable Choice

No strong unit heterogeneity

Units are similar enough that a stable, unmeasured trait isn't driving results.

You want the population-average relationship

Not the within-unit effect, just the overall association across the sample.

A useful, honest baseline

Before adding fixed or random effects, pooled OLS shows what a naive model would say.

The Standard-Error Problem

Even when $\hat{\beta}$ is fine, classical OLS standard errors assume every observation is independent. In a panel, that is almost never true.

Why: α_i alone creates serial correlation

Even if ε_{it} is i.i.d., two periods of the same unit share α_i:

$$\text{Corr}(u_{it}, u_{is}) = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_\varepsilon^2} > 0, \quad t \neq s$$

And shocks often persist: a firm's bad year echoes into the next.

The consequence

Ignoring within-unit correlation **understates** standard errors $\Rightarrow$ inflated t-statistics $\Rightarrow$ you risk declaring a result "significant" when the uncertainty is much larger.

Petersen (2009) shows how common — and how large — this problem is in finance panels.

The Fix: Cluster-Robust Standard Errors

Clustering by unit *i* lets errors correlate *arbitrarily* within a unit over time, while assuming independence across units.

$$\widehat{V}_{CR}(\hat{\beta}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\Big(\sum_{i=1}^{N} \boldsymbol{X}_i'\hat{\boldsymbol{u}}_i\hat{\boldsymbol{u}}_i'\boldsymbol{X}_i\Big)(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

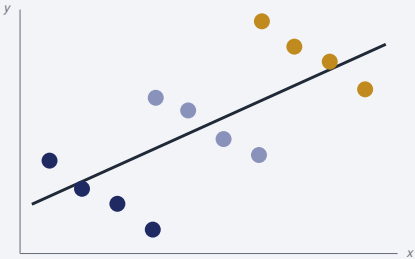
1. **Point estimates unchanged** — Clustering corrects inference, not the coefficients
2. **SEs usually adjust upward** — Often substantially, once within-unit correlation is accounted for
3. **Cluster on the unit** — Two-way (unit and time) if you also suspect common shocks across units
4. **You need many clusters** — Theory relies on $N \to \infty$; with few clusters, consider the wild cluster bootstrap

Cameron & Miller (2015) is the standard practitioner's guide.

What a Single Pooled Line Can Hide

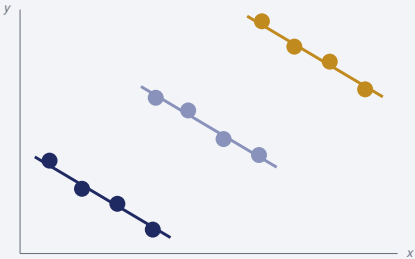
If units differ systematically, one fitted line can average away — or even **reverse** — the true relationship (Simpson's paradox).

What pooled OLS sees



One line, positive slope

What's really happening per unit



Three units, each with a negative slope

Same 12 data points in both panels. Units with higher α_i also have higher x: exactly the case $\text{Cov}(x_{it}, \alpha_i) \neq 0$.

Testing Poolability: The F-Test

Before trusting the pooled model, test whether unit-specific intercepts are jointly equal.

H_0: $\alpha_1 = \alpha_2 = \cdots = \alpha_N$ (poolable) vs. H_1: at least one α_i differs

Compare the pooled model's residual sum of squares with that of a model including unit dummies (LSDV):

$$F = \frac{(SSR_{POLS} - SSR_{LSDV})/(N-1)}{SSR_{LSDV}/(NT-N-K)} \sim F_{N-1,\, NT-N-K} \quad \text{under } H_0$$

Reject $H_0 \Rightarrow$ pooling is not supported; move to fixed or random effects.

Stata reports this test automatically at the bottom of `xtreg, fe` output (next class).

If the Poolability Test Fails

1	**Unit-specific effects are real**	Some stable, unobserved trait drives part of the variation in *y*
2	**A single pooled line is misspecified**	It averages away meaningful differences across units
3	**The next step is choosing an estimator**	Fixed or random effects, decided next class with the Hausman test

Reading the Pooled OLS Output

$\hat{\beta}_1$ in pooled OLS answers one specific question:

> *"Across all units and all periods pooled together, how does y move, on average, with x?"*

Important: it does not distinguish between

- **Between-unit variation:** comparing one firm with another
- **Within-unit variation:** comparing a firm with its own past

Pooled OLS uses both at once, and the weight each gets depends on the data — not on your research question.

Between vs. Within Variation

$$x_{it} = \underbrace{\bar{x}_i}_{\text{between}} + \underbrace{(x_{it} - \bar{x}_i)}_{\text{within}}, \qquad \bar{x}_i = \tfrac{1}{T_i} \textstyle\sum_t x_{it}$$

Between variation

- Differences across units, averaged over time
- *Why is Firm A more profitable than Firm B overall?*
- Vulnerable to unit-level confounders

Within variation

- Differences for the same unit, across time
- *Why is Firm A more profitable this year than last?*
- Nets out anything time-invariant about the unit

Pooled OLS blends both into a single coefficient — it can't tell you which one drives the result. `xtsum` *shows you how much of each your data has.*

The Upside: Degrees of Freedom and Efficiency

$N \times T$

observations to estimate from, far more than a single cross-section

$\downarrow$

sampling variance of $\hat{\beta}$, all else equal

$\uparrow$

power to detect economically meaningful effects

But: efficiency is only valuable if the estimate is unbiased in the first place. A precise but biased estimate is not progress.

Pooled OLS in Stata (1/2): Declare and Inspect the Panel

```
clear all
webuse nlswork          // NLS of Young Women, aged 14-26 in 1968
xtset idcode year       // unit = idcode (each woman), time = year
xtdescribe              // balanced? how many periods per unit?
xtsum ln_wage hours     // overall, between, and within variation
```

`xtset`	declares the panel structure. Pooled OLS does not use it, but every `xt` command (including `xtreg`) does
`xtdescribe`	shows participation patterns: a quick check of how unbalanced the panel is
`xtsum`	splits each variable's standard deviation into between and within components

Pooled OLS in Stata (2/2): Estimate and Compare SEs

```
regress ln_wage hours union age                        // classical SEs
estimates store naive
regress ln_wage hours union age, vce(cluster idcode)   // by woman
estimates store clustered
estimates table naive clustered, b se                  // same b, new SEs
```

`regress`	ordinary least squares on the stacked $N \times T$ sample
`vce(cluster idcode)`	cluster-robust SEs, allowing correlation within each woman over time
`estimates table`	places both models side by side: coefficients are identical, standard errors are not

In class: run both and compare. How much do the SEs on `union` change?

Pooled OLS: Strengths and Limitations

Strengths

+ Simple, transparent, easy to communicate
+ Uses every available observation ($N \times T$)
+ The benchmark for other models (e.g., the Breusch–Pagan LM test against random effects)
+ Fine when heterogeneity truly is negligible

Limitations

— Biased if unit effects correlate with x

— Naive SEs overstate confidence

— Blurs between and within variation

— Can hide or reverse the true relationship

Check Your Understanding

1	**Clustering**	You estimate firm ROA on board independence by pooled OLS (2,000 firms $\times$ 10 years) and cluster by firm. Which problem does clustering fix — and which does it *not* fix?
2	**Serial correlation**	Why are composite errors of the same firm correlated over time even if ε_{it} is i.i.d.?
3	**Simpson's paradox**	A colleague's pooled slope is positive, but every firm's own slope is negative. What does this tell you about $\text{Cov}(x_{it}, \alpha_i)$?
4	**Poolability**	The F-test rejects H_0 with $p < 0.001$. What do you report next?

Recap and What's Next

Takeaway: pooled OLS is the simplest panel estimator and a useful baseline — but it is only trustworthy when unobserved unit effects are uncorrelated with your regressors, and only reliable once standard errors are clustered.

Next class: Fixed vs. Random Effects

How to model unit heterogeneity directly instead of hoping it doesn't matter — then the Breusch–Pagan and Hausman tests to choose between models.

Questions?

References

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